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Gibbs-BERTopic: A Hybrid Approach for Short Text Topic Modeling (February 2025)

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ABSTRACT As a rich source of direct user needs, online reviews can be effectively analyzed through topic modeling to uncover user preferences and requirements. However, the short and unstructured nature of online reviews, along with their high dimensionality, noise, and complex semantic structure, poses challenges for traditional topic modeling techniques. The topic modeling technique BERTopic utilizes high-dimensional embeddings to capture text semantic information, but it faces the problems of sparsity, noise, and uncertainty when dealing with small-scale data. To reduce information barriers and improve the robustness and accuracy of topic models, we propose a new enhanced topic model—a hybrid method that integrates Gibbs sampling with BERTopic (Gibbs-BERTopic). Gibbs-BERTopic builds on BERTopic and iteratively optimizes and adjusts the topic distribution by Gibbs sampling the initial topic modeling results of BERTopic to improve the quality of the topic model. Experimental results show that Gibbs-BERTopic significantly outperforms the original BERTopic model, achieving over a 20% increase in silhouette coefficient and nearly a 20% improvement in topic coherence. It is verified that Gibbs-BERTopic can effectively reduce the impact of noise and data sparsity on topic extraction, optimize topic distribution, enhance topic clustering effect, and thus extract topics that are more in line with the actual data characteristics and potential structure.

INDEX TERMS Topic modeling, online reviews, short text, BERTopic, Gibbs sampling, statistical method, natural language processing, data mining.

I. INTRODUCTION

With the rapid advancement of information technology, the explosive growth of text data has made it increasingly crucial to extract valuable thematic information from it [1]. Topic modeling, as one of the most effective methods in the field of text mining, plays a pivotal role in understanding user needs and preferences [2]. However, online reviews, characterized as unstructured short texts, pose unique challenges for text processing and topic analysis due to their high sparsity, high dimensionality, large volume, and complex grammatical and semantic structures [3, 4]. Conventional LDA(Latent Dirichlet Allocation) topic models, which rely on implicitly capturing document-level word co-occurrence patterns to unveil topics, are severely affected by data sparsity in short documents [5] and are better suited for long texts [6, 7]. Additionally, they exhibit certain limitations when handling large-scale text data, primarily due to their inadequate ability to capture contextual information and limited capacity to process sparse and noisy data [8]. To address these shortcomings, various deep learning-based topic models have emerged in recent years, aiming to better capture semantic information from text [9].

BERTopic (Bidirectional Encoder Representations from Transformers for Topic Modeling) [10], a topic modeling approach that leverages BERT(Bidirectional Encoder Representations from Transformers) embeddings and clustering algorithms, captures contextual semantic information from text through high-dimensional embeddings, generating more accurate and fine-grained topic representations. Unlike traditional topic modeling techniques, BERTopic utilizes BERT's powerful contextual word embedding technique to capture the semantics and context of words in the corpus, making it particularly well-suited for short texts with low information density and limited contextual cues [11]. The model fully utilizes the advantages of pre-trained language models in encoding text in advanced natural language processing techniques, solving common phrase dependency and semantic ambiguity problems in traditional topic models[11]. BERTopic consistently exhibits superior performance when handling large-scale datasets. Owing to the powerful representation capabilities of SBERT(Sentence-BERT) and the high efficiency of UMAP(Uniform Manifold Approximation and Projection), BERTopic can generate topics rapidly, accurately, and stably on large-scale datasets [12].

Despite its effectiveness, BERTopic also exhibits certain limitations. Firstly, when applied to small datasets (fewer than 1000 documents), BERTopic may encounter overfitting issues due to data sparsity and suboptimal model training. This is because the BERT model struggles to learn and capture sufficient semantic information from small datasets, and BERTopic relies on a substantial amount of data to better represent topic semantics and distribution [13, 14]. Secondly, BERTopic restricts each document to a single topic, precluding the extraction of topic distribution information for individual documents [10]. Finally, BERTopic may generate more outliers than anticipated in certain scenarios. Overall, BERTopic's shortcomings in handling data sparsity, noise, and small datasets can lead to an excessive number of topics and numerous outliers [15]. These issues can cause topic distribution instability and hinder modeling performance.

Gibbs sampling, a probabilistic approximate inference method, is particularly well-suited for high-dimensional models compared to other approximate inference approaches. It iteratively converges to the model's posterior distribution, providing a more accurate estimate of the topic distribution [16]. Gibbs sampling is characterized by its simplicity, efficiency, and accuracy. As an unbiased method, it not only recovers the true posterior distribution but also enables sparse updates since each token is associated with only one topic rather than the distribution over all topics, rendering it more efficient [17]. Gibbs sampling effectively adjusts the document-topic distribution, mitigating the effects of long-tail distributions, sparsity, and noise, thereby enhancing the stability and robustness of topic modeling. Gibbs sampling optimization is mainly used to optimize the topic distribution in topic models, especially in topic modeling frameworks like BERTopic, where it can be used to adjust the contribution of each document to each topic to enhance the stability and robustness of the model. Additionally, Gibbs sampling optimization can help balance the relationship between documents and topics, thereby improving the model's fit to text data.

To address the aforementioned limitations and enhance the effectiveness of topic modeling, this study proposes a hybrid approach that integrates BERTopic with Gibbs sampling. Initially, BERTopic is employed for preliminary topic modeling. Subsequently, Gibbs sampling is utilized to iteratively refine the initial topic representations. The optimized topic results are then fed back into the BERTopic model for dimensionality reduction and topic distribution updates. Finally, the Gibbs-BERTopic model is evaluated and the results are analyzed. This hybrid model significantly outperforms the original topic model.

The Gibbs-BERTopic hybrid model offers several advantages over the standalone BERTopic model:

1） Enhance the ability to capture topic distribution

This study introduces a novel extension to BERTopic by integrating Gibbs sampling, which enables a more granular optimization of the topic distribution. This enhancement significantly improves BERTopic's capacity to capture the fine-grained thematic structure of the data and generate more precise topic probability distributions for each document.

2） Deal with data sparsity

Gibbs sampling exhibits strong robustness in statistical modeling, providing more stable topic models in the presence of sparse data. When combined with BERTopic's semantic representation and dimensionality reduction clustering capabilities, this hybrid approach is expected to excel in handling sparse datasets.

3） Reduce outliers and excessive themes

This approach incorporates Gibbs sampling to achieve a better-balanced topic distribution during model training, reducing the impact of noisy data on topic generation. This effectively minimizes the occurrence of outliers and an excessive number of topics.

Therefore, the combination of BERTopic and Gibbs sampling is a justifiable and flexible approach, capable of adapting to diverse domains and text types.

This study introduces a novel hybrid modeling approach that leverages the strengths of deep learning representations, dimensionality reduction, and statistical sampling. By harnessing the semantic representation capabilities of BERT embeddings and the statistical optimization power of Gibbs sampling, the method effectively improves topic modeling performance, especially when dealing with short texts. The synergistic combination of these techniques allows the model to adapt to the characteristics of short text data and overcome the limitations of traditional topic modeling methods.

In this study, online reviews from Shanghai Haichang Ocean Park were used as a dataset to train the models. Comparing the performance of the Gibbs-BERTopic model with the baseline BERTopic model revealed that Gibbs-BERTopic achieved higher silhouette coefficients(ss) and topic coherence scores. Experiments demonstrate that Gibbs-BERTopic effectively reduces noise and topic uncertainty, leading to significant improvements in the robustness of topic modeling. These findings further demonstrate the advantages of Gibbs-BERTopic in handling data scarcity and generating more interpretable topics. The primary contributions of this research include:

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1) This study introduces a novel hybrid model, Gibbs-BERTopic, by integrating BERTopic with Gibbs sampling. This synergistic approach leverages BERTopic's strong text embedding capabilities and the probabilistic optimization of Gibbs sampling, resulting in a more robust and effective topic modeling framework.
2) The proposed hybrid model demonstrates effectiveness across datasets of varying scales. By optimizing topic probability distributions and leveraging a two-layer BERTopic structure, the model exhibits improved robustness in handling noise and sparsity.
3) Gibbs-BERTopic significantly outperforms the traditional BERTopic model in terms of topic coherence and clustering quality, particularly for short text datasets. By integrating Gibbs sampling, the model enhances semantic information capture and improves overall topic modeling performance.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 presents the overall framework of our proposed research, providing a detailed algorithm for Gibbs-BERTopic. Section 4 describes the experimental setup, including dataset selection, parameter settings, and evaluation metrics. Section 5 presents a comparative analysis of the experimental results. Section 6 discusses the advantages and potential applications of Gibbs-BERTopic. Finally, Section 7 concludes the paper and discusses future research directions.

II. RELATED WORKS

A. TOPIC MODELING

The advent of the data-intensive era has rendered traditional data processing and analysis methods inadequate. Consequently, text mining has emerged as a crucial approach for extracting information from textual data. It involves transforming unstructured text data into meaningful and functional information [18]. Among the various methods of text mining, topic modeling has gained significant attention. Topic modeling is a technique used to extract latent topics from text data to analyze its structure and identify potential themes, thereby revealing deeper semantic information [19].

In the traditional natural language processing domain, latent Dirichlet allocation (LDA), a common topic model, assumes that each document is composed of multiple topics, and each topic is represented by a probability distribution over words. Through iterative algorithms, LDA can obtain the distributions of topics and words [20]. LDA is efficient in handling large-scale text data. However, since reviews are often short and unstructured texts containing only a few words [21], LDA may suffer from sparsity issues and fail to obtain reliable topic distributions. BTM(Biterm Topic Model) addresses the problem of short text length and semantic sparsity by modeling word pair co-occurrence [5]. Nevertheless, BTM primarily focuses on the co-occurrence of word pairs and may neglect semantic information in larger contexts. In the field of deep learning for natural language processing, models such as LSTM(Long Short-Term Memory), Transformer, BERT, and their variants have been employed for topic extraction. LSTM, a classic model for sequential data, can capture contextual information within text, aiding in understanding word relationships and enhancing topic extraction performance [22]. Referencing [22], a topic-enhanced LSTM model was proposed to address document representation issues. Evaluations on real-world datasets for document classification, topic detection, information retrieval, and document clustering demonstrated the superiority of the topic-enhanced LSTM model over baseline models. Beyond these methods, recent advancements in topic modeling have introduced novel techniques, such as Top2Vec, which employs vector embeddings and clustering to discover topics in semantic space [23]. However, Top2Vec exhibits shortcomings in generating semantically related and document-relevant keywords [24]. In contrast, BERTopic leverages the powerful capabilities of the BERT language model for text encoding and representation [7].

B. BERTOPIC AND GIBBS SAMPLING

Traditional topic models often neglect the contextual information of words within sentences. To address this issue, Maarten Grootendorst developed BERTopic in 2022 [10], a modeling approach that creates continuous topic representations through clustering techniques and class-based TF-IDF. By leveraging the BERT model, text data is transformed into high-dimensional embedding vectors. BERT's ability to capture contextual information enables the generation of semantically rich embeddings [25]. According to reference [4], BERTopic can capture coherent language patterns in short texts and consistently outperforms GSDMM. In terms of improving the BERTopic framework, reference [26] proposed an optimized framework by iteratively finding the optimal silhouette coefficient through min_cluster_size to enhance topic coherence. Furthermore, reference [27] integrated a dynamic hyperparameter optimizer into the existing BERTopic framework to learn the optimal dimensionality of different document embeddings, developing an automated pipeline that can provide more coherent and diverse topic sets using moderate-sized datasets.

BERTopic has proven particularly effective for topic modeling in short texts, leading to its widespread adoption across various domains. In the finance domain, for instance, a comparative study [28] of three state-of-the-art topic models—Latent Dirichlet Allocation (LDA), Top2Vec, and BERTopic—revealed that BERTopic outperformed its counterparts in analyzing the impact of financial market news. It required minimal data preprocessing, achieved the highest coherence scores, exhibited the best interpretability, and demonstrated a reasonable computational efficiency. However, BERTopic's lack of an inherent smoothing mechanism renders it susceptible to suboptimal performance when confronted with sparse data, noise, or small sample sizes. This can lead to increased instability in topic distributions during the modeling process, thereby compromising the effectiveness of topic clustering.

Gibbs sampling, through iterative updates, can smooth both the document-topic and topic-word distributions, gradually optimizing topic assignments and mitigating the impact of sparse data. As a Markov chain Monte Carlo (MCMC) method, Gibbs sampling is employed to draw samples from complex probability distributions. In topic modeling, the Gibbs sampling algorithm is utilized to infer the topic distribution of each document and the word distribution of each topic. By iteratively updating these distributions, the Gibbs sampling algorithm can gradually approximate the true topic and word distributions, providing a more accurate thematic interpretation of the text data [29]. According to [17], a key advantage of Gibbs sampling is its unbiased nature, allowing it to converge to the true posterior distribution given an infinite number of iterations. Additionally, Gibbs sampling enables sparse updates, as each token is associated with only one topic, rather than a distribution over all topics. This makes Gibbs sampling more efficient when handling large-scale datasets. As demonstrated in [30], this paper utilizes Gibbs sampling to infer the unknown variables in LDA, including the document-topic distribution and the topic-word distribution. Through an iterative sampling process, these latent variables are estimated, revealing the dominant topics within documents and identifying the words most strongly associated with each topic. Furthermore, [31] constructs a topic model using Gibbs sampling. By optimizing parameters, generating topics, and extracting labels on Arabic and English datasets, it derives the optimal parameter combinations, topic lists, labels, and corresponding evaluation results (such as coherence scores) for each dataset. Consequently, Gibbs sampling is widely regarded as an effective approach for learning topic models.

Therefore, this paper proposes a Gibbs-BERTopic approach. Initially, BERTopic is employed to conduct a preliminary topic modeling. Subsequently, Gibbs sampling is utilized to iteratively refine the initial topic representations, reducing noise and obtaining more robust topic distributions. The optimized results are then fed back into the BERTopic model to enhance the quality of topic clustering and improve the robustness of the topic model.

III. GIBBS-BERTOPIC METHODOLOGY

A. OVERVIEW OF THE PROPOSED GIBBS-BERTOPIC MODEL

User reviews, as unstructured short texts, provide valuable insights into real-world user preferences and needs. However, short text data is often sparse and noisy, which poses a challenge for traditional topic modeling techniques. To address these challenges, this study proposes Gibbs-BERTopic, a hybrid probabilistic topic modeling approach that integrates the contextual semantic representation capabilities of BERTopic with the statistical optimization strength of Gibbs sampling. The research framework of this paper is shown in Figure 1, which mainly consists of five key parts:

1) Data Collection and Preprocessing: User reviews of Shanghai Haichang Ocean Park were scraped from China Ctrip Travel Online (Ctrip.com). The data underwent preprocessing, including Jieba segmentation, stop word removal, synonym replacement, custom word loading, and filtering words shorter than two characters.

2) Document Embedding, Dimensionality Reduction, Clustering, and Topic Modeling: The preprocessed text data was embedded using UMAP for dimensionality reduction and clustered using HDBSCAN(Hierarchical Density-Based Spatial Clustering of Applications with Noise). This process resulted in an initial topic distribution, which was generated by the BERTopic model.

3) Topic Optimization via Gibbs Sampling: An initial topic distribution is obtained via BERTopic, and then Gibbs Sampling is applied to iteratively optimize the document-topic probability distribution. Through multiple iterations, the document-topic assignments and topic-word distributions are refined, gradually converging to an optimal topic distribution.

4) BERTopic Model Refinement: After the Gibbs sampling optimization, the refined topic distribution is fed back into the BERTopic model. This allows for further UMAP dimensionality reduction and an update of the topic distribution, ensuring the model's accuracy.

5) Model Evaluation and Analysis: The final Gibbs-BERTopic model is evaluated using relevant metrics, and the results are analyzed to assess the model's performance.

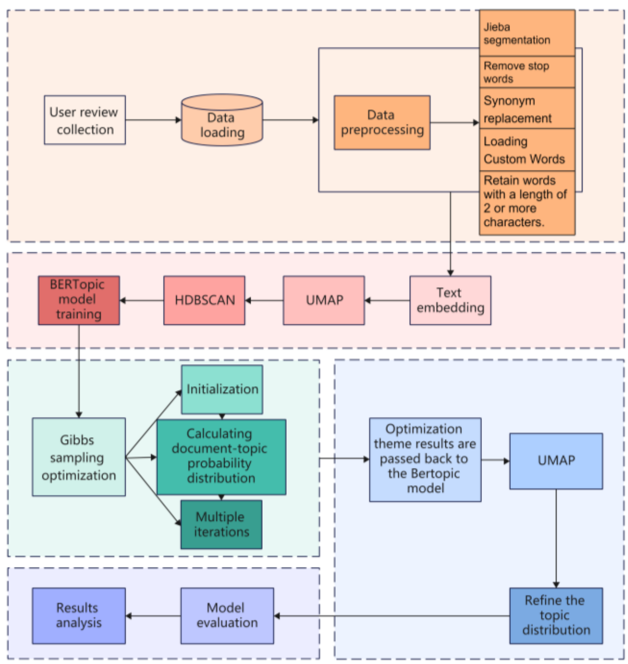

FIGURE 1. **Research Framework.**

The proposed framework addresses key challenges of short text topic modeling, including data sparsity, noise, and

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excessive topic generation, thereby enhancing the robustness, stability, and interpretability of the topic model. The framework consists of three main phases: Initial Topic Generation using BERTopic, Gibbs Sampling Optimization, Feedback to BERTopic and Final Optimization to achieve refined topic distributions. The following sections describe these three steps in detail.

B. INITIAL TOPIC GENERATION USING BERTOPIC

The first phase employs BERTopic to generate an initial topic distribution. BERTopic uses BERT embeddings to encode high-dimensional semantic representations of the text. The workflow includes:

1) Text Embedding: Preprocessed text is transformed into vector representations using Sentence-BERT (SBERT), which preserves contextual and semantic information.

2) Dimensionality Reduction: To reduce computational complexity, UMAP (Uniform Manifold Approximation and Projection) is applied to project the embeddings into a lower-dimensional space.

3) Clustering: Using HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), documents are grouped into topic clusters, and an initial set of topics T is generated.

4) Topic-Word Distribution Initialization: The topic-word distribution ϕ is initialized as: $\phi_k \sim Dirichlet\ (\beta)$, where k is the topic index, and β is the hyperparameter controlling word distribution sparsity.

5) Document-Topic Distribution: The prior topic distribution π and hyperparameter α are used to initialize the document-topic distribution θ, influencing the topic assignment Z.

At the end of Phase I, BERTopic provides an initial topic assignment Z, document-topic distribution θ, and topic-word distribution ϕ.

C. GIBBS SAMPLING OPTIMIZATION

To refine the initial topic distributions, the Gibbs sampling phase is introduced for probabilistic optimization. The process is as follows:

1) Topic Assignment Updates: For each word W_{dn} in document d, the topic assignment $Z_{dn}^{(t+1)}$ is updated using the posterior probability:

$$P(Z_{dn}^{(t+1)} = k \mid \boldsymbol{Z}^{(t)}, \boldsymbol{W}, \alpha, \beta) \propto \frac{n_{d,k}^{(t)}+\alpha}{n_{d,}^{(t)}+K\alpha} \cdot \frac{n_{k,w_{dn}}^{(t)}+\beta}{n_{k,+}^{(t)}+V\beta} \tag{1}$$

Where $n_{d,k}^{(t)}$ denotes the number of words in document d that belong to topic k at the t-th iteration, and $n_{k,w_{dn}}^{(t)}$ denotes the number of times word W_{dn} appears in topic k at the t-th iteration. Where k is the Total number of topics and V is the Vocabulary size.

2) Topic-Word Distribution Updates: Simultaneously, the topic-word distribution ϕ_k is updated as shown in (2).

$$\phi_k^{(t+1)} = \frac{n_k+\beta}{n_{k,}+V\beta} \tag{2}$$

where n_k denotes the term frequency of each word in topic k, $n_{k,}$ indicates the total word count for topic k, and V represents the vocabulary size.

3) Document-Topic Distribution Updates： The document-topic distribution θ_d is updated according to (3).

$$\theta_d^{(t+1)} = \frac{n_d+\alpha}{n_{d,}+K\alpha} \tag{3}$$

Here, n_d represents the number of words assigned to each topic in document d, and $n_{d,}$ denotes the total word count in document d.

4) Iteration and Convergence: After each iteration, the topic-word distribution ϕ is updated using a normalized count of words assigned to each topic, and the document-topic distribution θ is updated based on the topic proportions in each document. Gibbs sampling iteratively updates Z, ϕ, and θ until convergence, where the topic distributions stabilize.

Gibbs sampling iteratively updates the document-topic distribution and topic-word distribution, gradually converging to a more stable topic distribution. However, recognizing the potential uncertainty introduced by the Gibbs sampling process, several measures were implemented to ensure the stability and reliability of the model. First, we conducted multiple independent Gibbs sampling runs on each dataset to reduce the influence of randomness from a single sampling process. Second, during each iteration, we dynamically monitored the changes in the log-likelihood to ensure that the sampling process was converging to a stable topic distribution. Finally, after completing the sampling, the resulting topic distributions were manually validated for semantic consistency, integrating domain knowledge to ensure that the extracted topics accurately reflected user needs and adhered to semantic logic. Through these strategies, we were able to significantly reduce the uncertainty introduced by Gibbs sampling, thus enhancing both the stability and interpretability of the model's results.

D. FEEDBACK TO BERTOPIC AND FINAL OPTIMIZATION

Once the Gibbs sampling process stabilizes, the refined topic distributions are fed back into BERTopic for further dimensionality reduction and topic clustering: The updated document-topic distribution θ is used as input to UMAP for dimensionality reduction. HDBSCAN is re-applied to refine the topic clustering. The topic-word distribution ϕ is used to regenerate and evaluate the final topics. This iterative optimization process ensures that the final topic assignments are robust, stable, and semantically coherent.

IV. EXPERIMENTAL DESIGN

A. DATA COLLECTION AND PREPROCESSING

In this study, a web crawler was employed to collect 4,260 reviews of Shanghai Haichang Ocean Park from Ctrip, a major Chinese travel platform. To address the challenges of BERTopic when dealing with long-tail distributions, data sparsity, noise, and small-scale data, and to optimize the performance of topic modeling, this study divided the data into two datasets: a large dataset comprising all collected reviews (4,260 in total) and a smaller dataset consisting of 658 consecutive reviews with tem-poral characteristics, extracted from the complete dataset. Both datasets were cleaned to remove duplicates.

During the data preprocessing stage, this study customized a stop word list by adding meaningless words to the Harbin Institute of Technology stop word list [32], reducing the probability of low-semantic words appearing in the topic results and significantly saving time compared to creating a stop word list from scratch. This study removed stop words, non-textual components, and punctuation, replaced synonyms, loaded custom words, and retained words with a length of at least two characters. Jieba, a widely adopted and efficient Chinese word segmentation tool [33], offers functionalities such as word segmentation and keyword extraction. In this study, the open-source Jieba library was utilized to segment continuous text into individual words, facilitating subsequent text processing and experimental analysis.

B. PARAMETER SETTINGS

This study employed BERTopic's default embeddings, which are based on Sentence Transformers, to generate text embeddings as vector representations of the text. Sentence Transformers create document embeddings from a set of documents and typically have a sentence length limitation of 512 characters, making them suitable for shorter texts. BERTopic utilizes pre-trained semantic models to compute embedding vectors for each document, capturing the semantic representation of the document in a vector space.

This study applied UMAP to reduce the dimensionality of the sentence embeddings created in the previous step. First, it reduces computational complexity and memory usage, leading to more efficient processing. Second, it facilitates better clustering with HDBSCAN, an algorithm that performs more effectively in low-dimensional spaces. Additionally, the process of manifold learning inherent in UMAP can reveal more local semantic features within the data. UMAP is currently considered one of the most effective methods for dimensionality reduction of text vectors. The key parameters in UMAP are n_neighbors, n_components, and metric. In this study, the same values were set for both UMAP steps within BERTopic. Specifically, n_neighbors was set to 15. This choice allows for better preservation of local structure information by considering a larger number of neighboring data points. This, in turn, helps to accurately map data points in the high-dimensional space. n_components was set to 2, reducing the embeddings to a two-dimensional space for visualization purposes. This setting also strikes a balance between capturing essential information and achieving dimensionality reduction. Finally, the metric parameter was set to cosine. This metric utilizes cosine similarity to measure the distance between data points, effectively capturing semantic similarity. This makes it particularly well-suited for analyzing natural language text data.

Given that UMAP preserves some of the original high-dimensional structure, HDBSCAN, with its robustness and efficiency, outperforms traditional clustering algorithms in identifying dense clusters (i.e., popular topics) and is more resilient to noise in the dataset [34]. Therefore, this study employs HDBSCAN for topic clustering and adopts a dynamic optimizer to iteratively adjust the min_cluster_size (mcs) parameter within the range of 5 to 45. Smaller values of mcs are suitable for detecting small or rare clusters in the data, enhancing the capture of fine-grained details but potentially introducing more noise or meaningless clusters. Conversely, larger values of mcs help filter out the noise and small clusters, improving the overall stability and reliability of the clustering results, but may overlook some small but important data structures. Experiments show that setting min_cluster_size to 10 can better achieve the goals of cluster analysis and enhance the understanding of data structures and patterns, depending on specific data characteristics and analysis requirements.

By setting the calculate_probabilities parameter of BERTopic to True, the model is enabled to compute the probability distribution of topics, allowing for the quantification of each document's contribution to various topics. This significantly enhances the interpretability and comprehensibility of the topic model. To optimize the topic distribution and document-topic probability distribution, the Gibbs sampling iterations were set to 5. Multiple iterations help stabilize and improve the accuracy and consistency of the topic model, especially when dealing with complex datasets. However, Gibbs-BERTopic generated an excessive number of topics on the dataset, as shown in Figure 2. Many topics exhibited overlap or intersection due to high cosine similarity, indicating that the model's topic number was not optimal. An excessive number of topics can lead to over-dispersion of topics within the target domain, thereby diminishing the significance of each topic. Therefore, this study adopted a manual approach to reduce the number of topics, aiming to achieve a topic partition that enables semantic differentiation. Based on the visualized categories in Figure 2, we set k to 5 to reduce model complexity, improve interpretability, and enhance efficiency while maintaining analytical performance. Therefore, considering the experimental results and model performance evaluation, this study selected the following hyperparameters: n_neighbors=15, n_components=2, metric=cosine, min_cluster_size=10, iterations=5, and k=5 to develop a robust hybrid model that excels at capturing fine-grained details, producing superior topic distributions, filtering noise, and demonstrating stability on complex datasets.

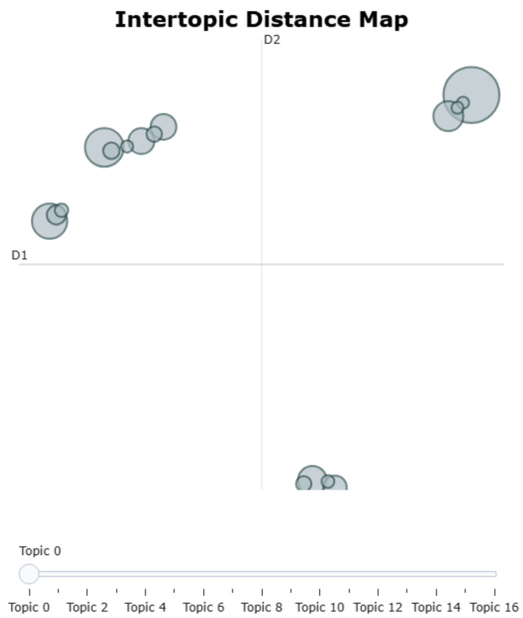

FIGURE 2. **Topic Clusters Automatically Generated Using Gibbs-BERTopic.**

C. MODEL EVALUATION METRICS

The evaluation of short text topic models is an ongoing research area, and the choice of evaluation metrics depends heavily on the specific dataset [35]. While various metrics have been proposed, including topic coherence [36, 37], this study adopts a more comprehensive approach by considering both clustering and topic coherence perspectives to evaluate short text topic models.

To validate the effectiveness of the Gibbs-BERTopic model, this study compares Gibbs-BERTopic with a single BERTopic model. Silhouette coefficient and topic coherence are employed as evaluation metrics to assess the performance of both models.

The silhouette coefficient is a crucial metric for evaluating clustering quality [38]. It measures the clarity of clusters after clustering, with higher values indicating better clustering performance. The formula is shown in (4) to (7).

$$ss = \frac{1}{N}\sum_{i=1}^{N} ss\,(i) \tag{4}$$

$$ss(i) = \frac{b(i)-a(i)}{max(a(i),b(i))} \tag{5}$$

$$a(i) = \frac{1}{n-1}\sum_{j\neq 1}^{n} distance\ (i,j)\ \ i \in A \tag{6}$$

$$b(i) = \frac{1}{n}\sum_{j=1}^{n} distance\ (i,j)\ \ i \in A \neq C, j \in C \tag{7}$$

In this context, $a(i)$ represents the average distance between sample i and other samples within the same cluster, as shown in (6). Meanwhile, $b(i)$ denotes the average distance between sample i and the samples in the nearest neighboring cluster, as illustrated in (7).

Topic coherence is a commonly used metric to evaluate the effectiveness of topic models. It is a cohesion measure calculated for individual topics to assess the consistency and validity of the topics generated by the topic model [36]. Higher topic coherence indicates stronger correlations and better interpretability of words within a topic. This study utilizes the Gensim library to compute the coherence score for each topic. Gensim supports the calculation of several topic coherence metrics, including c_v, c_uci, and c_npmi. c_v coherence is an evaluation metric based on word co-occurrence statistics to measure the consistency of words within a topic and is applicable in most situations. c_uci coherence employs a statistical method based on word frequency and word distribution to assess topic coherence. c_npmi coherence utilizes a variant of Pointwise Mutual Information (PMI) to measure topic consistency. Reference [26] explores the Spearman correlation(also known as Spearman's rank correlation coefficient) between silhouette coefficient and topic coherence, using different min_cluster_size (mcs) to find the optimal silhouette coefficient to improve topic coherence. This paper analyzes the correlation between silhouette coefficient and various topic coherence metrics under different mcs using Spearman correlation analysis. Based on the Spearman correlation results obtained from large and small-scale datasets, the best topic coherence metric is selected to evaluate the quality of the topic model. By iterating mcs from 5 to 45, the paper calculates the silhouette coefficient and scores of three topic coherence metrics for the Gibbs-BERTopic model on two different sized datasets, and explores the relationship between the silhouette coefficient(ss) and the three metrics(c_v, c_uci, c_npmi) using Spearman correlation. The Spearman correlation coefficients are shown in Tables 1, 2, and 3.

Table 1. **Spearman Correlation Coefficients between ss and c_v.**

	Small_ss	Small_c_v	All_ss	All_c_v
Small_ss	1(0.000)	-0.812(0.008)	0	0
Small_c_v	-0.812(0.008)	1(0.000)	0	0
All_ss	0	0	1(0.000)	-0.7(0.036)
All_c_v	0	0	-0.7(0.036)	1(0.000)

Table 2. **Spearman Correlation Coefficients between ss and c_uci.**

	Small_ss	Small_c_uci	All_ss	All_c_uci
Small_ss	1(0.000)	0.276(0.472)	0	0
Small_c_uci	0.276(0.472)	1(0.000)	0	0
All_ss	0	0	1(0.000)	0.25(0.516)
All_c_uci	0	0	0.25(0.516)	1(0.000)

Table 3. **Spearman Correlation Coefficients between ss and c_npmi.**

	Small_ss	Small_c_npmi	All_ss	All_c_npmi
Small_ss	1(0.000)	0.377(0.318)	0	0
Small_c_npmi	0.377(0.318)	1(0.000)	0	0
All_ss	0	0	1(0.000)	0.483(0.187)
All_c_npmi	0	0	0.483(0.187)	1(0.000)

According to the Spearman correlation analysis results, there is a significant negative correlation between the two variables ss and c_v in both datasets, with correlation coefficients of -0.812 and -0.7, respectively. This indicates that the relationship between the two variables is close and negative. In addition, the p-values of 0.008 and 0.036 are both less than 0.01 and 0.05, respectively, indicating that this negative correlation is statistically extremely significant, reaching the 1% and 5% significance levels. In the ss and c_uci correlation coefficient table, since p is 0.472 and 0.516, which are greater than the commonly used significance levels of 0.05 or 0.01, this correlation is not statistically significant. Therefore, the null hypothesis cannot be rejected, that is, there is no real correlation between ss and c_uci. This trend may be due to random error or sample specificity. In the ss and c_npmi correlation coefficient table, since the p-values are 0.318 and 0.187, which are greater than the usual significance level thresholds of 0.05 or 0.01, the null hypothesis cannot be rejected, and it cannot be considered that there is a statistically significant correlation between ss and c_npmi. Additionally, the self-correlation of both variables is 1, indicating that each observation is perfectly correlated with itself, as expected.

The above analysis reveals a significant negative correlation between ss and c_v. This implies that as the ss value increases, the c_v value tends to decrease; conversely, when the ss value decreases, the c_v value tends to increase. This relationship is statistically highly reliable and warrants further investigation into the underlying mechanisms of mutual influence between the two variables. Given the significant correlation between the silhouette coefficient and c_v, and the consistently higher coherence scores achieved by c_v across different mcs values compared to the other two metrics, this paper selects c_v as the preferred metric for evaluating topic coherence.

V. RESULTS ANALYSIS

A. SILHOUETTE COEFFICIENT COMPARISION

This section delves into a comparative analysis of the silhouette coefficient (ss) results obtained for both models, Gibbs-BERTopic and single BERTopic, across different iterations of the min-cluster-size (mcs) parameter on the two datasets. The mcs parameter plays a crucial role in determining the minimum size of clusters within the clustering process. Its setting significantly impacts the topic model, influencing the number and diversity of generated topics. Smaller mcs values lead to the generation of more topics. This is because a smaller mcs allows clustering fewer documents into a single cluster, resulting in a higher number of generated topics. However, this may also increase topic redundancy as similar documents might be assigned to different clusters. Conversely, larger mcs values lead to the generation of fewer topics. This is because a larger mcs requires clustering more documents into a single cluster, thereby reducing the number of generated topics. Similar documents might be assigned to the same cluster, potentially decreasing topic diversity. Therefore, selecting an appropriate mcs value is crucial. By adjusting mcs, the number and diversity of generated topics can be controlled to achieve better topic modeling results, tailored to the specific dataset and task requirements [26]. Table 4 presents a comparison of the silhouette coefficients (ss) for Gibbs-BERTopic and the single BERTopic model. It is evident that regardless of the mcs choice, the Gibbs-BERTopic model consistently outperforms the single BERTopic model in terms of silhouette coefficient values on both datasets. Notably, Gibbs-BERTopic achieves silhouette coefficient values exceeding 70% for the small-scale dataset, demonstrating consistent and superior performance compared to the large-scale dataset. A high silhouette coefficient generally indicates tighter clusters with better separation, suggesting that Gibbs sampling optimization can further refine the initial clustering results obtained by BERTopic, enhancing clustering accuracy and compactness, leading to superior clustering outcomes. This characteristic is particularly pronounced for the small-scale dataset.

The results of this study indicate that Gibbs-BERTopic outperforms the standard BERTopic algorithm in terms of silhouette coefficient values for small-scale datasets. This superior performance indicates that Gibbs-BERTopic exhibits a more robust ability to partition topics, leading to the generation of clearer and more meaningful topics. Consequently, Gibbs-BERTopic produces topics with enhanced inter-topic distinction and intra-topic coherence. Furthermore, Gibbs-BERTopic exhibits greater adaptability and stability across different parameter settings when applied to small-scale datasets. This characteristic translates into more consistent model performance.

Table 4. **Silhouette Coefficient Comparison.**

mcs	Small_BERTopic_ss	Small_Gibbs-BERTopic_ss	All_BERTopic_ss	All_Gibbs-BERTopic_ss
5	0.112437	0.842141	-0.18529	0.468221
10	0.037021	0.747931	-0.09894	0.473251
15	0.033738	0.736724	0.356968	0.713726
20	0.274393	0.742638	0.25785	0.687134
25	0.631567	0.886083	0.345758	0.656615
30	0.641762	0.881911	0.38308	0.636403
35	0.661586	0.885253	0.305854	0.623938
40	0.700549	0.88774	0.229411	0.694226
45	0.705465	0.88749	0.400877	0.703933

B. TOPIC COHERENCE COMPARISON

Topic coherence metrics assess the degree of semantic relatedness within a topic. In the previous section of this study, the c_v metric was selected to evaluate topic coherence. The relationship between ss and c_v values for the dataset was explored, revealing a significant negative correlation. The line graph depicting this relationship, with mcs on the x-axis, is shown in Figure 3. Based on the observed proximity between ss and c_v values, mcs = 10 was chosen to compare the topic coherence of Gibbs-BERTopic and standard BERTopic for small-scale datasets. On the small-scale dataset, BERTopic achieves a topic coherence score of 0.62623, while Gibbs-BERTopic attains a significantly higher score of 0.813084, representing a substantial improvement of nearly 20% in topic coherence, as shown in Table 5. This remarkable enhancement in topic coherence underscores the superior performance of Gibbs-BERTopic compared to single BERTopic, particularly in the context of small-scale datasets. The c_v metric directly reflects the coherence and interpretability of topics. Higher c_v values generally indicate stronger semantic relatedness among the words within a topic, leading to better explanations of the topic's meaning and content. It can also be seen from the table that an examination of the topic coherence scores for the large-scale dataset reveals a similar trend to that observed for the small-scale dataset. In this study, the c_v values obtained for both models on both datasets further corroborate the superior topic coherence of Gibbs-BERTopic. This advantage is particularly evident for the small-scale dataset, demonstrating the model's ability to effectively handle smaller datasets.

The results demonstrate that Gibbs-BERTopic significantly outperforms the conventional BERTopic model on small datasets, exhibiting higher topic coherence. The words generated by Gibbs-BERTopic exhibit strong semantic relatedness, enabling more accurate topic interpretation. These findings highlight the remarkable improvements brought by the Gibbs sampling method in topic modeling tasks, particularly in enhancing topic coherence and quality.

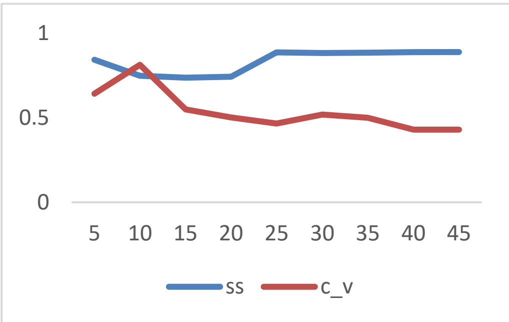

FIGURE 3. **Line plot of ss and c_v values for small datasets.**

Table 5. **Comparison of topic coherence between Gibbs-BERTopic and BERTopic.**

Model	Small_ss	Small_c_v	All_ss	All_c_v
BERTopic	0.037021	0.62623	-0.09894	0.544818
Gibbs-BERTopic	0.747931	0.813084	0.473251	0.786907

C. RUNTIME AND MEMORY USAGE COMPARISON

This section analyzes the runtime and memory usage of the proposed Gibbs-BERTopic. Gibbs-BERTopic employs a two-layer BERTopic framework with embedded Gibbs sampling to optimize topic distribution. However, Gibbs sampling has the drawback of high computational cost, requiring O(K) time complexity per sampling step, leading to increased computational expense [39]. Consequently, the Gibbs-BERTopic model consumes more runtime and memory compared to the single BERTopic model, as shown in Figures 4 and 5, where time is measured in seconds and memory is measured in GB.

The experiments were conducted on a computer with the following specifications: 1) Processor: AMD Ryzen 7 5800H with Radeon Graphics @3.20 GHz. 2) Memory: 16GB. 3) Operating system: Windows 11. All parameters were set to their default values.

As shown in Figure 4, Gibbs-BERTopic consumes more runtime than the single BERTopic model on small datasets, but the increase is still reasonable. Despite employing a two-layer BERTopic framework and Gibbs sampling for optimization, Gibbs-BERTopic's runtime is only 30% higher than that of the single BERTopic model. Figure 5 illustrates that both models consume over 1.8GB of memory on small datasets. Gibbs-BERTopic's memory usage is slightly higher than that of the single BERTopic model, with a difference of less than 200MB. While Gibbs-BERTopic incurs higher runtime and memory usage, the additional time is limited and stable. Moreover, Gibbs-BERTopic achieves superior topic quality compared to the single BERTopic model (refer to the previous section).

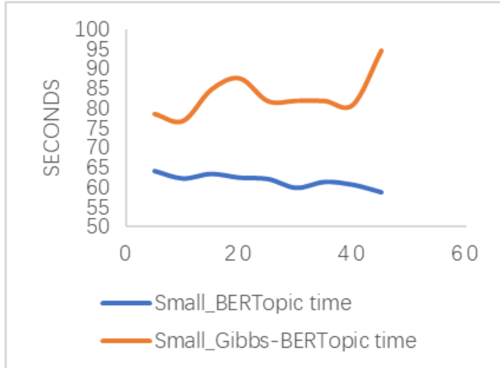

FIGURE 4. **Comparison of model runtime on small datasets.**

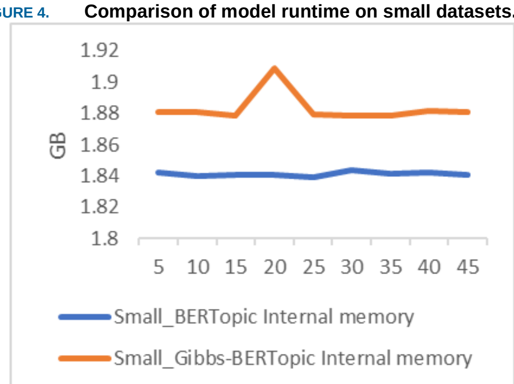

FIGURE 5. **Memory usage comparison of models on small datasets.**

D. TOPIC VISUALIZATION

Topic visualizations are generated using Gibbs-BERTopic under a small dataset. In the section, Six automatically generated topics were selected, with Topic 0 having the highest number of reviews (414), followed by Topic 1 (65), Topic 2 (34), Topic 3 (27), Topic 4 (24), and Topic 5 (20). The generated topic word visualizations are shown in Figure 6. To construct the visualizations, the c-tf-idf method was employed to merge all documents within a class into a single document. Next, tf-idf was applied to compute the top n keywords for each class [40]. Finally, the topic words within each class were ranked from highest to lowest based on their c-tf-idf scores. The topic words with the highest scores are considered the most representative of their respective topics. C-tf-idf scores reflect the importance of each keyword within the generated topics, indicating the extent to which each word represents a particular topic, thereby enhancing the interpretability of the generated topics [28]. As shown in Figure 6, the top topic words for each class are "children," "very good," " killer whale show," "marine life," "poor value for money," and "too many people."

Topic 0 revolves around the target audience, as evident from the associated keywords: "children," "holidays," "children like it very much," "suitable for parent-child," and "second visit." These keywords highlight the popularity of Shanghai Haichang Ocean Park among children, making it an ideal destination for family outings during vacations. This finding suggests that businesses should prioritize children as their tar-get audience and tailor their marketing strategies to promote parent-child activities. By incorporating interactive elements designed for both children and parents, businesses can enhance visitor satisfaction and improve the overall park experience.

Topic 1 focuses on the positive sentiment expressed by visitors, as evidenced by the associated keywords: "very good," "very fun," "quite fun," "awesome," and "fantastic." These keywords reflect the overwhelmingly positive evaluations of Shanghai Haichang Ocean Park among visitors. Additionally, the relatively close scores of the top five keywords indicate a consistent level of satisfaction across the park's offerings. This finding underscores Shanghai Haichang Ocean Park's remarkable achievement in garnering high levels of overall visitor satisfaction. These insights hold immense value for businesses aiming to enhance their customer experience and attract a wider visitor base. Businesses can ensure that attractions, facilities, and amenities are well-maintained, safe, and up-to-date, and continuously strive to enhance the visitor experience by addressing customer needs and expectations to foster a culture of customer satisfaction, enhance their reputation, and attract a loyal customer base.

Topic 2 explores the amusement programs that left a lasting impression on visitors, as evident in the associated keywords: "killer whale show," "roller coaster," "overall design," "not too big," and "five stars." The prominence of "killer whale show" and "roller coaster" highlights these attractions as iconic elements of the Shanghai Haichang Ocean Park experience. Additionally, the presence of "not too big" and "overall design" suggests that visitors are not only seeking thrilling rides but also value overall design and spatial considerations. This finding underscores the importance of striking a balance between exciting rides and thoughtful design in creating a memorable visitor experience. Businesses should prioritize the development of high-quality attractions that cater to a variety of preferences while also ensuring that the overall park layout and design are aesthetically pleasing and conducive to a positive visitor experience.

Topic 3 delves into the marine life and themed elements that resonated with visitors, as evidenced by the associated keywords: "marine life," "polar bear," "mermaid," "aquarium," and "killer whale show." These keywords underscore the strong alignment of these elements with the marine theme of Shanghai Haichang Ocean Park. This finding suggests that businesses can further enhance the visitor experience and promote marine conservation by incorporating more immersive marine-themed elements into their operations. By creating engaging and educational experiences that showcase the wonders of the marine world, businesses can not only entertain visitors but also instill a deeper appreciation for marine life and conservation efforts.

Topic 4 focuses on the negative sentiment expressed by visitors, as reflected in the associated keywords: " poor value for money," "not many people," "not very good," "not fun," and " the fly in the ointment." These keywords provide valuable insights into areas where Shanghai Haichang Ocean Park can improve its offerings and enhance the overall visitor experience. By carefully analyzing negative feedback, businesses can identify specific issues that are negatively impacting visitor satisfaction. This feedback can serve as a roadmap for improvement, guiding businesses in addressing shortcomings and enhancing the overall quality of their products, services, operations, and customer experience.

Topic 5 delves into experience-related focus, encompassing keywords such as " too many people," "average experience," "roadside stalls," "rushed atmosphere," and "salespeople." The data reveals that visitors are primarily concerned with park congestion, overall experience, amenities, recreational activities, and service quality. Understanding these experience concerns empowers managers to gain valuable insights into visitor behavior, preferences, and needs. This information proves invaluable for crafting personalized marketing strategies and refining experience design. Experience focus serves as a cornerstone for managers to elevate product and service quality, optimize marketing strategies, enhance brand reputation, and ultimately, boost customer satisfaction.

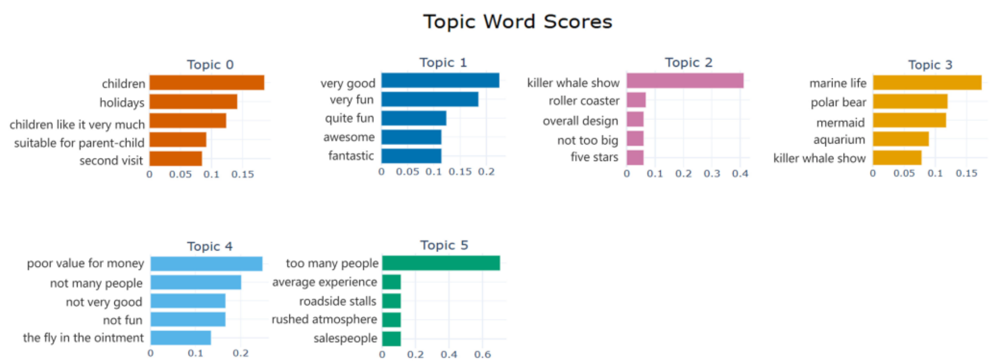

FIGURE 6. **Thematic Bar Chart.**

VI. DISCUSSION

Topic modeling, as a cornerstone technology in text mining and information retrieval, has garnered substantial attention in recent years [41]. This paper explores a hybrid model combining BERTopic and Gibbs sampling, aiming to enhance the robustness of topic models when confronted with sparse, noisy, and small-scale data. By mitigating the impact of noise, sparse data, and improving semantic understanding and topic coherence, our proposed method leverages the text encoding capabilities of BERT and the optimization power of Gibbs sampling. Compared to traditional topic modeling approaches, this method offers significant advantages and innovations.

The proposed hybrid topic modeling approach combining BERTopic and Gibbs sampling exhibits several key advantages. Firstly, by leveraging BERT's ability to extract semantic information from text, it can not only capture lexical-level correlations but also understand contextual and semantic relationships within the text, thereby enhancing the depth of topic modeling in comprehending textual content. Secondly, Gibbs sampling optimizes the topic distribution and document-topic probability distribution, improving the model's performance in terms of topic coherence and separability, resulting in more interpretable and practical topics. This significantly enhances the effectiveness of topic modeling in real-world applications. Compared to traditional LDA methods, the proposed method demonstrates significant advantages. Traditional LDA, based on the bag-of-words model, has limitations in understanding semantic information [35]. In contrast, BERTopic employs BERT to encode text, enabling more accurate representation of semantic information and improving the accuracy and coherence of topic modeling. Furthermore, Gibbs sampling optimization further refines the topic distribution, resulting in more interpretable and practical topics. The proposed method is potentially more suitable for handling text data with strong semantic correlations compared to traditional LDA.

In practical applications, this improved topic modeling approach exhibits broad prospects. For instance, in information retrieval and content analysis, it can assist users in rapidly and accurately understanding the thematic structure of large-scale text data, optimizing information retrieval performance. In academic research, it can help identify research frontiers and hotspots in the literature, providing powerful tool support for the academic community. The proposed method is particularly suitable for scenarios that require deep semantic understanding and topic analysis of text. It can be applied to various fields such as text mining, market research, and sentiment analysis, helping users quickly obtain key information and structured representations from text data. However, it is necessary to pay attention to the impact of data quality during application, especially for data in specific domains or languages. Effective data preprocessing and corpus construction are required to ensure the model's effectiveness and stability.

VII. CONCLUSIONS

To address the challenges of online review short texts, such as short length, sparse semantics, noise, and small-scale data, as well as the limitations of existing models, this paper proposes Gibbs-BERTopic. The model first conducts a preliminary training of BERTopic using preprocessed data. Subsequently, the initially generated topics are iteratively optimized through Gibbs sampling. Finally, the optimized topic model is fed back to the BERTopic model for dimensionality reduction and updating the topic distribution. The study uses user online reviews of Shanghai Haichang Ocean Park as the data source for topic modeling, extracting adjacent data as a small-scale dataset, and all data as a large-scale dataset. The results demonstrate that the Gibbs-BERTopic model, utilizing Gibbs sampling and a two-layer BERTopic structure, outperforms the single BERTopic model in terms of silhouette coefficient and topic coherence, while also exhibiting favorable performance in terms of running time and memory usage.

While the BERTopic-Gibbs sampling hybrid model offers numerous advantages, there are still some considerations and limitations. Firstly, large-scale datasets may require more computational resources and time to train the model and optimize parameters, especially considering the high computational cost of Gibbs sampling. Secondly, the model's ability to handle long and short texts needs further optimization and adjustment to accommodate text data of varying lengths and complexities.

Future research can further refine Gibbs-BERTopic to address more complex and diverse user review data. The model can also be applied to other domains, such as market research and sentiment analysis, to help enterprises explore higher-value user needs and thus provide a basis for enterprise decision-making. Future research directions can also continue to explore issues such as multimodal integration, improved interpretability, and large-scale data processing to promote the further development of topic modeling in practical applications.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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